

Introduction to NLP

What is Natural Language
Processing?

Natural Language Processing

- how to program computers to *process* and analyze large amounts of *natural language* data

Information Extraction

Subject: **curriculum meeting**

Date: January 15, 2012

To: Dan Jur

Event: Curriculum mtg
Date: Jan-16-2012
Start: 10:00am
End: 11:30am
Where: Gates 159

Hi Dan, we've now scheduled the curriculum meeting.

It will be in Gates 159 tomorrow from 10:00-11:30.

-Chris

Create new Calendar entry

Information Extraction & Sentiment Analysis



Attributes:

zoom
affordability
size and weight
flash
ease of use

Size and weight



- ✓ • nice and compact to carry!
- ✓ • since the camera is small and light, it is easy to carry around those heavy, bulky cameras either!
- ✗ • the camera feels flimsy, is plastic and very light in weight you have to be very delicate in the handling of this camera

Machine Translation

- Fully automatic
- Helping human translators

Enter Source Text:

这 不过 是 一 个 时 间 的 问 题 .

Translation from Stanford's *Phrasal*:

This is only a matter of time.

Enter Source Text:

تعرض الرئيس اللبناني اميل لحود لـ حملة عنيفة في مجلس النواب الذي انعقد امس في جلسة تشريعية عاجية تحولت الي " محاكمة " لـ رئيس الجمهورية علي موقفه من المحكمة الدولية و " الملاحظات " التي ادلي بها حول هذا الموضوع .

Translate

Clear

Enter Translation:

lebanese

president

suffered

exposed

president emile

before

presented

Done!

offer

Language Technology

making good progress

mostly solved

Spam detection

Let's go to Agra!



Buy Fake Painiting ...



Part-of-speech (POS) tagging

ADJ ADJ NOUN VERB ADV

Colorless green ideas sleep furiously.

Named entity recognition (NER)

PERSON ORG LOC

Einstein met with UN officials in Princeton

Sentiment analysis

Best roast chicken in San Francisco!



The waiter ignored us for 20 minutes.



Coreference resolution

Carter told Mubarak he shouldn't run again.

Word sense disambiguation (WSD)

I need new batteries for my *mouse*.



Parsing

I can see Alcatraz from the window!

Machine translation (MT)

第13届上海国际电影节开幕...



The 13th Shanghai International Film Festival...

Information extraction (IE)

You're invited to our dinner party, Friday May 27 at 8:30



Party
May 27
add

still really hard

Question answering (QA)

Q. How effective is ibuprofen in reducing fever in patients with acute febrile illness?

Paraphrase

XYZ acquired ABC yesterday

ABC has been taken over by XYZ

Summarization

The Dow Jones is up

The S&P500 jumped

Housing prices rose



Economy is good

Dialog

Where is Citizen Kane playing in SF?

Castro Theatre at 7:30. Do you want a ticket?





Ambiguity makes NLP hard: “Crash blossoms”

Violinist Linked to JAL Crash Blossoms

[Violinist Linked to JAL Crash] Blossoms

[Violinist] Linked to [JAL Crash Blossoms]

Main
Verb

Teacher Strikes Idle Kids

[Teacher Strikes] Idle [Kids]

[Teacher] Strikes [Idle Kids]

Ambiguity is pervasive

New York Times headline (17 May 2000)

Fed raises interest rates

Fed raises interest rates 0.5%

Why else is natural language understanding difficult?

non-standard English

Great job @justinbieber! Were SOO PROUD of what youve accomplished! U taught us 2 #neversaynever & you yourself should never give up either♥

segmentation issues

the New York-New Haven Railroad
the New York-New Haven Railroad

idioms

dark horse
get cold feet
lose face
throw in the towel

neologisms

unfriend
Retweet
bromance

world knowledge

Mary and Sue are sisters.
Mary and Sue are mothers.

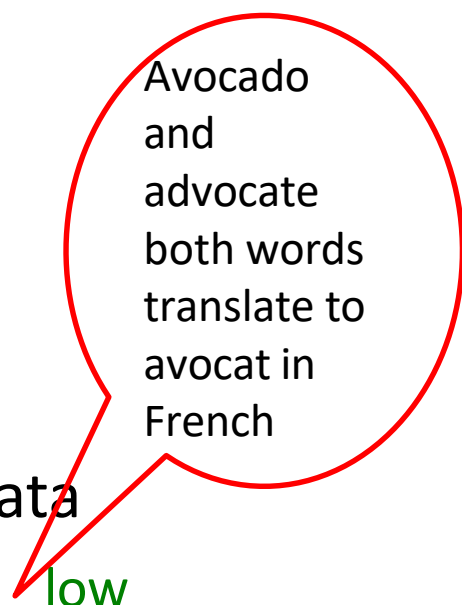
tricky entity names

Where is *A Bug's Life* playing ...
Let It Be was recorded ...
... a mutation on the *for* gene ...

But that's what makes it fun!

Making progress on this problem...

- The task is difficult! What tools do we need?
 - Knowledge about language
 - Knowledge about the world
 - A way to combine knowledge sources
- How we generally do this:
 - probabilistic models built from language data
 - $P(\text{"L'avocat général"} \rightarrow \text{"the general avocado"})$ **low**
 - $P(\text{"L'avocat général"} \rightarrow \text{"the general advocate"})$ **high**
 - Luckily, rough text features can often do half the job.



Avocado
and
advocate
both words
translate to
avocat in
French

NLP-related Resources

- [NLTK](#)(Python-based)
- [Keras, TensorFlow \(Deep Learning\)](#)
- [Statistical natural language processing and corpus-based computational linguistics: An annotated list of resources](#)
- [Stanford NLP parser](#) (Stanford University NLP group)
- [OpenNLP](#) (Apache)
- [LingPipe](#) (Jave-based)

Top Conferences and Journals in NLP Field

Conferences

- ACL(The Association for Computational Linguistics)
<https://www.aclweb.org>
- HLT-NAACL
- EMNLP
- COLING

Journals

- ACM Transactions on Speech and Language Processing
- Computational Linguistics
- International Journal of Computational Linguistics
- Information Processing and Management (Journal)
- Knowledge and Data Engineering (Journal)
- Information Science (Journal)
- Knowledge Based systems (Journal)

Machine Learning Toolkits

- [Weka](#) (A rich collection of machine learning algorithms, Machine Learning Group at the University of Waikato)
- [Mallet](#) (An alternative package for Weka, developed by Andrew McCallum at University of Massachusetts Amherst)
- [LibSVM](#) (A collection of SVMs, developed by Chih-Chung Chang and Chih-Jen Lin at National Taiwan University)
- [SVM-light](#) (Another collection of SVMs, developed by Thorsten Joachims at Cornell University)
- [GraphLab](#) (Large-scale machine learning package)
- [mahout](#) (Apache large-scale machine learning package)
- [Topic Models](#) (David Blei's collection of various topic models)

Course Outline